Cardboard Automata Arcade

Build a one-crank machine that performs a mini task.

Win condition: Most reliable motion with useful mechanism and clear explanation.

THE SCIENCE YOU ARE USING

Cams, followers, linkages. Turning a crank spins a cam. A follower rides on the cam and rises and falls as the shape passes, turning steady rotation into up-and-down or back-and-forth motion. Linkages pass that motion along. The cam shape decides the movement.

Reliability. A machine that jams is a failed machine. Smooth holes, low friction, and a steady crank make motion repeatable. You build a hand-cranked automaton that completes a task and runs reliably, then explain the mechanism.

HOW TO BUILD AND RUN IT

01 **Build the box and shaft.** Make a sturdy box and pass a skewer through straw bearings so it turns freely.

02 **Cut a cam.** Cut a cam disc and fix it to the shaft. An off-center or oval cam gives more motion.

03 **Add a follower.** Rest a vertical rod on the cam so it rises and falls as you crank.

04 **Top it with a task.** Attach a character or tool that performs a mini task as the follower moves.

05 **Tune for reliability.** Reduce friction and wobble so it runs the same every turn.

Safety: Hot glue is staff-supervised. Scissors stay seated and closed when not in use. Allergy: None.

MATERIALS

Cardboard	shared
Skewers	shared
Straws	bearings
Craft sticks	shared
Tape	shared
Hot glue	staff station
Scissors	per team

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): ____________________

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The ONE variable we are testing: ____________________

Baseline result: ____________________

After redesign: ____________________

DEFEND YOUR DESIGN

What evidence made you change your design? ____________________

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If you ran it again, what is the ONE thing you would change? ____________________

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HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Reliable motion	30
Cam or linkage complexity	20
Task success	20
Explanation	20
Aesthetics and build quality	10
Total	**100**